

Math 4650 Final Project: Silly Walk Analysis

Composed by Brandon Bolar

Abstract

This Math 4650 statistical project analyzes an article published in the British Medical Journal: *"Quantifying the benefits of inefficient walking: Monty Python inspired laboratory-based experimental study."* Primarily, the energy expenditure (EE; kcal/kg/min) of two particular styles is compared to that of a normal walking type. In this project, it has been found that Usual Walking has a significant correlation with the Putey Walking form. The Teabag Walking form is statistically not strongly correlated with either the Usual or Putey Walking forms. BMI is an essential factor in examining the study's measurements. The statistical significance level for alpha for this analysis was set at $\alpha = 0.05$.

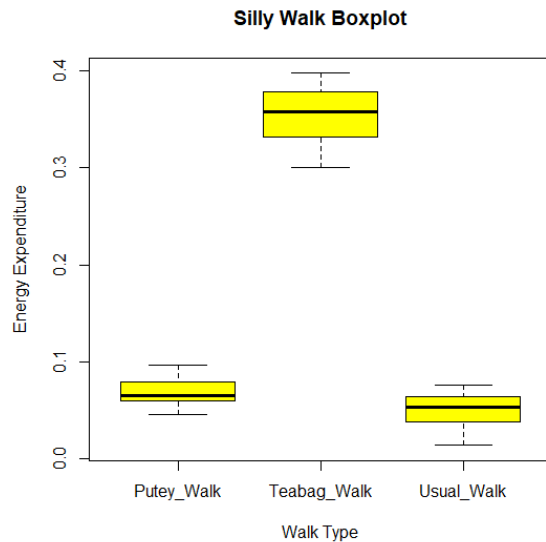
These research trials are needed for exploring non-orthodox movements that may be efficient based on factors such as gender, energy expenditures (EE; kcal/kg/min), and BMI (Body-Mass-Index). The result of the study used ANOVA, Tukey analysis, linear regression, QQ plots, and colored boxplots to determine this project's conclusion. These were dissected further in the Analysis sector of the report.

Introduction

The null hypothesis of these trials is that each of the three walking styles has no remarkable statistical differences from each other. The alternative hypothesis we will delve into is the analysis of significant differences between the three walking forms concerning each other if prompted by the evaluated data.

The data was collected by 13 separate trials for each of the three walking types and categorized by a subset of 7 identified males and 6 identified females. Gender was another variable to look at in the information collected in the study. The measurements were given, and the data was expected to be interpreted from this study. Each walk was quantified on a 30-meter course.

Analysis



Let's begin with the side-by-side boxplot revealing the median of each walk type. Putey Walk has a negative skew and a narrow mid-section that shows that 50% of its values are ranged close together. The Teabag Walk plot has a more normal distribution than the others. However, the data for Teabag Walk is of a higher magnitude in terms of its energy expenditure. This means it takes more kcal/kg/min to fulfill the motion of this walk. Teabag Walk has the greatest data range between values collected in the study. The Usual Walk's boxplot appears to have the lowest energy expenditure of the three walks. It has a slight negative skew and, due to its boxplot, has the most efficient walk. We will see later if the silly walks are strongly statically related by measuring their p-values from ANOVA.

Analysis of Variance Table

Response: efficiency

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
type	2	0.75284	0.37642	672.01	< 2.2e-16 ***
Residuals	36	0.02017	0.00056		

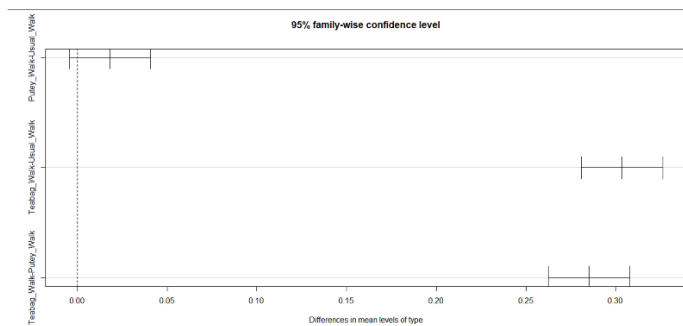
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

This ANOVA table produced by R tells us things about the data. Since the p-value is highly below the α , this breaks the null hypothesis. The idea that there is no significant difference between all the walk types is untrue. The large difference between the Teabag boxplots' medians and the other walks supports this notion.

Tukey multiple comparisons of means
95% family-wise confidence level
factor levels have been ordered

Fit: aov(formula = efficiency ~ type, data = onedata)

Stype		diff	lwr	upr	p adj
Putey_Walk-Usual_Walk	0.01819472	-0.004495873	0.04088531	0.1369009	
Teabag_Walk-Usual_Walk	0.30340647	0.280715878	0.32609706	0.0000000	
Teabag_Walk-Putey_Walk	0.28521175	0.262521159	0.30790234	0.0000000	



The Tukey's Honest Significance Test and its ordered graph show the relationship between one walk and another. P values between the Usual Walk and the Putey Walk are not significant when compared to each other. This is because it's above the chosen significance level. The relationship from any walk paired with the Teabag Walk has a p-value that, in fact, is larger than the α . The ordered graph is organized by increasing averages in the data, supports all these relationship conclusions.

```
Pairwise comparisons using t tests with pooled SD

data: onedata$efficiency and onedata$type

      Putey_Walk Teabag_Walk
Teabag_Walk <2e-16      -
Usual_Walk  0.17      <2e-16

P value adjustment method: bonferroni
```

Finally, we use the Bonferroni table to support the above statements. Usual and Putey Walks are not statistically different. Teabag Walk is significantly different from the other two walk types.

Analysis of Variance Table

```
Response: Teabag_Walk
      Df    Sum Sq   Mean Sq F value    Pr(>F)
BMI      1 0.0083938 0.0083938 20.1944 0.001154 **
data$gender 1 0.0010033 0.0010033  2.4137 0.151326
Residuals 10 0.0041565 0.0004156
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

Call:
lm(formula = Teabag_Walk ~ BMI + data$gender, data = mydata)

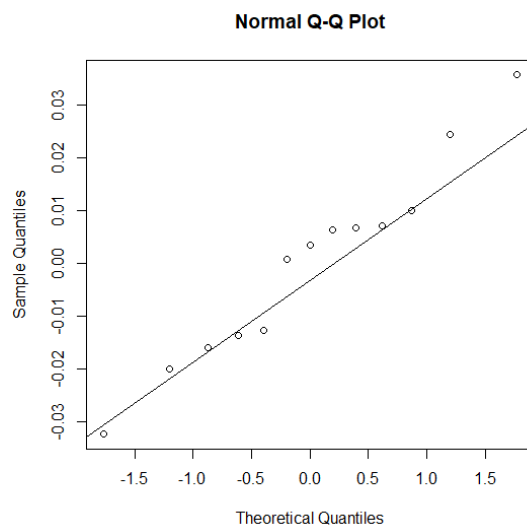
Residuals:
    Min       1Q   Median       3Q      Max
-0.032312 -0.013699  0.003413  0.007208  0.035670

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
BMI         0.007524   0.001601   4.701  0.00084 ***
data$genderfemale 0.170981   0.038727   4.415  0.00130 **
data$gendermale  0.151077   0.044451   3.399  0.00679 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

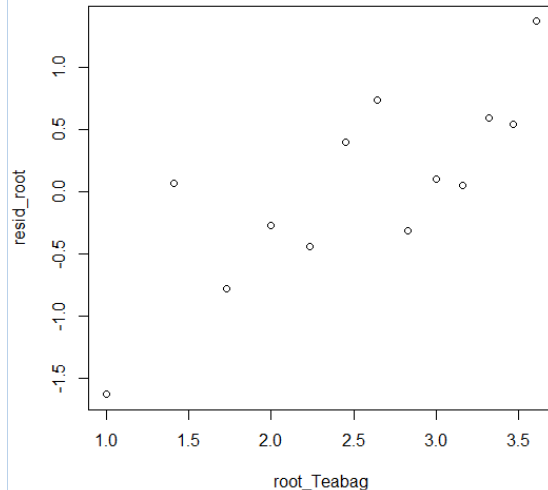
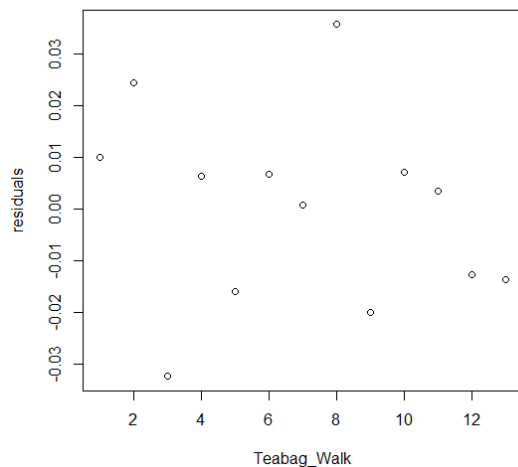
Residual standard error: 0.02039 on 10 degrees of freedom
Multiple R-squared:  0.9975,    Adjusted R-squared:  0.9967
F-statistic: 1308 on 3 and 10 DF,  p-value: 2.874e-13

```

Above is the ANOVA of Teabag Walk, BMI, and gender. Below that table is the linear regression of Teabag Walk, BMI, and Gender. The ANOVA highlights the strong relation of Teabag Walking to the BMI of those tested. The evidence is the p-score lower than the α . Gender p-score hovers below α at .13 % for females and .679% for males. Both p-scores are below α : 0.05, but It also demonstrates that it's more of an objective factor for the female gender. Pay attention to the R^2 calculation, which is closely approximated to .99 by the linear regression. This signifies that all the predictor components can describe almost all of the variance given by the model. Also, the observations and predicted quantities measured are unbiased because of R^2 .



The QQ Plot of Teabag Walk compared to BMI + Gender shows a strong linear correlation when graphed. This means there's a solid normal distribution of the gathered data despite the outliers at the ends of the line. The outlier values don't appear to be that extreme.



The right least squares transformation checks if the variance in the left residual graph is at a minimum. It has a line of regression that linearly fits the Teabag data nicely. It supports the idea that the Teabag Walk values forecasts will show dependency in the future.

Conclusion

What has been found in this study is that Silly Walking, like Teabag Walking, has more significant energy expenditure than Usual and Putey Walking. Still, its presence causes us to reject the null hypothesis that all walking types won't deviate from each other statistically. Since Usual Walking has the boxplot lowest mean, it is the most efficient way of walking if you have to choose one. This makes sense because it's the control group. Putey Walking ranks first among the Silly Walking techniques to conserve energy expenditure. Statistically, because the data relation between Putey and Usual Walking is not significant from each other. A postulation can be explained by inferring that the motion of Putey Walking is close to the actions of Usual Walking. The most inefficient walking technique tested is the Teabag style of walking. These conclusions are demonstrated by the boxplots and are reinforced by the Bonferroni tables and TukeyMSD details.

Now, let's focus on the analysis and data interpretation regarding Teabag Walking. Teabag Walking scales statistically with BMI. The p-value correlation was calculated below the α value while in the critical region. This rejects a null hypothesis that there is no significance to BMI data points. Gender also doesn't play a role in energy expenditure information. It is concluded that p-scores for BMI are a stronger factor than gender because BMI has a lower p-score than both females and males in the linear regression analysis. It's a bad idea to categorize the data by gender and instead utilize BMI. Collectively ANOVA calculations give gender a p-value higher than the given α . The QQ plot shows the Teabag compilations and their dependency on BMI in the residuals. The linear regression line proves that point. Furthermore, the least squares transformation of residuals graphed with Teabag data reveals that both natures scale linearly and is the best estimator for future values. The forecast is that the more collected Teabag data, the larger it will show dependency to BMI's. This reflects that the higher your BDMI level translates into the greater energy expenditure.

To improve this study, a measurement for stride span should take place because of the limited 30-meter limit of silly walks. A large stride would be hypothesized to expend less energy than shorter stride spans.

Under the conditions of the study, you want the most efficient energy expenditure for Teabag Walking, have a low BMI. Putey Walking can also be classified as a higher energy expended Usual Walking than an inefficient Silly Walk. The null hypothesis holds only for Usual and Putey Walking styles.